

EXOSKELETON SENSOR & ACTUATOR CONTROL SYSTEM

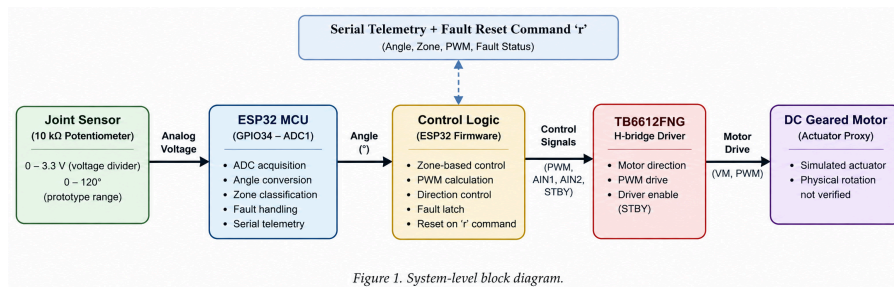
Embedded System, Biomedical Electronics and Power Systems - C-ROB TKMCE Internship Selection Task

1. Objective

Design and simulate a joint-angle sensing and safety-oriented actuator control architecture for a single-degree-of-freedom lower-limb exoskeleton joint using a potentiometer, ESP32 and TB6612FNG H-bridge stage. The design implements zone-based angle limiting with latched hard-fault protection and is verified in Wokwi simulation. This is a simulation/prototype architecture, not a medically approved or clinically validated device.

2. System Architecture

Signal path: Joint Sensor -> ESP32 ADC -> Angle Conversion -> Zone Classification -> Control Logic -> TB6612FNG -> DC Motor (actuator proxy), with serial telemetry and reset.



3. Sensor Interface and Angle Conversion

A 10 kOhm linear potentiometer is used as the joint-position sensor; its wiper connects to ESP32 GPIO34 (ADC1). The 12-bit ADC reading (0-4095) is mapped linearly to the 0-120 deg prototype angle range:

$$\text{jointAngle} = (\text{potRaw} / 4095.0) * 120.0$$

The 0-120 deg range is a simulation/prototype assumption, not a clinical range-of-motion specification. In Wokwi, the potentiometer is turned manually; it is not mechanically coupled to a physical joint or motor.

4. Control Logic and Operating Zones

The firmware classifies measured angle into five zones. LIMIT zones command proportional boundary-return motion; hard-limit zones latch a fault until serial command r is received.

Zone	Angle Range	Motor Action	Fault
FAULT_LOW	< 5 deg	Brake, latched	Yes
LIMIT_LOW	5-15 deg	Forward PWM (proportional)	No
SAFE	15-105 deg	Stop	No
LIMIT_HIGH	105-115 deg	Reverse PWM (proportional)	No
FAULT_HIGH	> 115 deg	Brake, latched	Yes

5. Actuator Interface - TB6612FNG

The driver stage receives PWM and direction commands from the ESP32. Physical motor rotation was not experimentally verified; the simulation verifies the command logic and reported driver-state outputs.

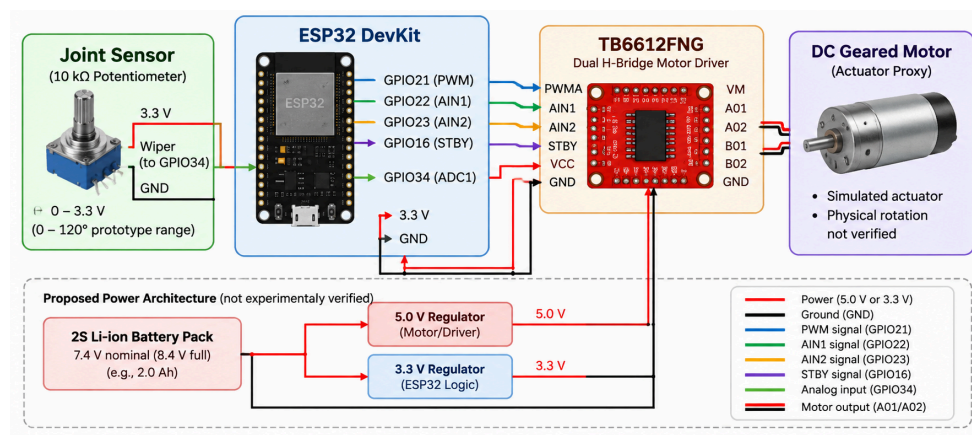


Figure 2. Circuit/interface diagram for the proposed actuator path.

Signal	ESP32 Pin	Function
PWM	GPIO21	Speed command
AIN1	GPIO22	Direction bit 1
AIN2	GPIO23	Direction bit 2
STBY	GPIO16	Driver enable (HIGH)

Truth table: STOP = AIN1=0, AIN2=0, PWM=0; FORWARD = AIN1=0, AIN2=1; REVERSE = AIN1=1, AIN2=0; BRAKE = AIN1=1, AIN2=1, PWM=255.

6. Safety and Proposed Power Architecture

Simulation-verified software safety: hard angle limits, latched FAULT_LOW/FAULT_HIGH states, brake-on-fault and manual serial reset. Proposed, untested hardware measures: normally-closed E-stop in the motor supply path, mechanical end-stops, PTC/fuse backup and local bulk/decoupling capacitance near the driver/motor.

Proposed power: For preliminary power-system sizing, a 2S Li-ion pack (7.4 V nominal, 8.4 V fully charged) is assumed. Separate regulated rails are proposed for the motor/driver and ESP32; the final motor-rail voltage must be selected from the actual actuator's rated voltage and current requirements. Example energy sizing from the existing design basis: $2.0 \text{ Ah} \times 7.4 \text{ V} = 14.8 \text{ Wh}$. The 5 W average draw is an assumption

used for preliminary sizing because motor current and mechanical load were not measured in simulation; therefore, the estimated runtime is $14.8/5 = 2.96$ h.

7. Wokwi Simulation and Test Evidence

The Wokwi implementation validates the sensor-to-control-to-driver command path by manually sweeping the potentiometer to emulate joint-angle input. The current simulation uses an ESP32, potentiometer and I/O indicators/scope for PWM and direction outputs; the physical actuator and mechanical sensor coupling are not simulated.

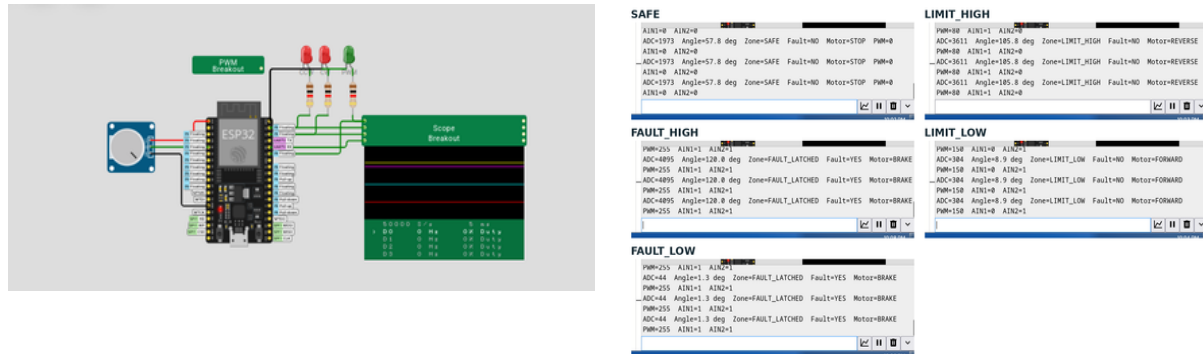


Figure 3. Wokwi circuit/interface diagram used for the simulation. The supplied project uses the ESP32, potentiometer, LED indicators and scope breakouts to represent the control I/O.

Test	ADC	Angle (deg)	Motor	PWM	AIN1/AIN2
SAFE	1973	57.8	STOP	0	0/0
LIMIT_HIGH	3611	105.8	REVERSE	80	1/0
FAULT_HIGH	4095	120.0	BRAKE	255	1/1
After reset (r)	-	57.8	STOP	0	0/0
LIMIT_LOW	304	8.9	FORWARD	150	0/1
FAULT_LOW	44	1.3	BRAKE	255	1/1

8. Limitations and Conclusion

Validation is simulation-only. The potentiometer and motor are not mechanically coupled in Wokwi; physical E-stop response, motor current, thermal behavior and mechanical end-stops remain untested. No clinical or medical validation is claimed. Before any human-worn or load-bearing prototype, the driver/motor electrical limits and independent hardware safety chain require bench validation.

Conclusion: the submitted controller demonstrates sensor interfacing, angle conversion, zone-based boundary-return commands, motor-driver command generation, latched fault handling and manual reset in simulation. Physical actuation and hardware protection remain future validation steps.

Simulation reference: [Wokwi project 474598962103266305](https://wokwi.com/project/474598962103266305).